

Estimating Yoga Benefits using Machine Learning and Multimodal Data

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Abstract—This paper presents a comprehensive, data-driven investigation into the physiological and psychological benefits of yoga, leveraging advanced machine learning techniques and multimodal biosignal data. The proposed system integrates five distinct types of biosensors—EEG, ECG, EMG, GSR, and SpO₂—implemented through custom-designed wearable hardware to capture detailed physiological responses. To facilitate robust model training, a hybrid dataset was constructed by combining the WESAD dataset, IEEE Health Sensor Data, and a Kaggle EEG-ECG Stress Dataset. These datasets were harmonized using transfer learning techniques and feature alignment strategies.

Following thorough preprocessing and feature engineering, we trained a deep learning-based regression model using a multi-layer perceptron (MLP) architecture to predict health improvements resulting from regular yoga practice. Experimental results demonstrate quantifiable enhancements in several physiological health indicators, thereby offering objective validation of yoga's benefits through biomedical signal analysis.

Keywords— Yoga, Machine Learning, Multimodal Data, Biomedical Sensors, Stress Detection, Deep Learning

I. INTRODUCTION

The pressures of modern life—characterized by hectic schedules, digital overload, and sedentary habits—have led to a significant rise in stress-related and lifestyle-induced health issues. In response, traditional wellness practices such as yoga have garnered global interest for their potential to restore balance, both physically and mentally. Yoga, with its rich combination of physical postures (asanas), controlled breathing techniques (pranayama), and meditative practices (dhyana), is widely believed to promote emotional resilience, stress reduction, and overall well-being. Despite the growing popularity and anecdotal support for yoga, scientific studies aiming to rigorously quantify its physiological effects remain limited, often relying on subjective reports and lacking reproducible, data-driven methods.

There is a growing recognition that to truly understand the impact of yoga, we need to move beyond personal testimonies and explore objective, measurable evidence. Recent

advancements in wearable biosensors and machine learning technologies offer a promising pathway. By capturing physiological signals before and after yoga sessions, and applying data-driven models to analyze these changes, we can begin to understand how yoga affects the human body in a quantifiable and personalized way. This approach not only lends credibility to the practice but also opens the door to adaptive and scalable applications in digital health and wellness.

In this study, we take a multimodal approach to data collection, utilizing a combination of physiological signals including electrocardiography (ECG), electroencephalography (EEG), electromyography (EMG), galvanic skin response (GSR), and blood oxygen saturation (SpO₂). Together, these signals provide a holistic view of the body's internal responses—ranging from heart activity and muscle tension to neural dynamics and stress levels—allowing us to capture subtle shifts in physical and mental states resulting from yoga.

Due to the lack of a single dataset containing all these modalities, we created a custom dataset by integrating data from several publicly available sources. The WESAD dataset served as our foundation for stress-related physiological data, which we enhanced with EEG and SpO₂ features sourced from the IEEE Health Sensor dataset and a Kaggle-based EEG-ECG dataset. This integration was carried out using feature engineering and harmonization techniques to ensure compatibility and consistency.

Once the dataset was constructed, we applied machine learning models to uncover patterns that might be missed by traditional statistical methods. These models help us understand which physiological markers respond most strongly to yoga and how these responses vary across individuals. By training the model on pre- and post-session data, we aimed to predict the effect of yoga on different biometric indicators with both accuracy and interpretability.

Ultimately, our goal is to bridge the gap between ancient wellness traditions and modern data science. By offering a method that is both rigorous and reproducible, we hope to

contribute to a more nuanced understanding of yoga's benefits. This work sets the stage for personalized, real-time wellness recommendations that adapt to an individual's physiological state—making yoga not only accessible but also measurable in meaningful ways.

II. RELATED WORK

A variety of studies have explored the detection of stress and evaluation of meditative states using physiological signals and machine learning techniques. These works collectively provide important insights and lay the foundation for data-driven wellness assessment systems like ours.

One study developed an explainable deep neural network (DNN) for stress detection using biosignals and handcrafted features extracted from the WESAD dataset [1]. Their model demonstrated an impressive accuracy of 96.57%, validating the potential of combining deep learning with human-engineered features in a multimodal sensor environment [9].

Another significant contribution analyzed heart rate variability (HRV) signals to distinguish between meditative and pre-meditative states. This study used empirical mode decomposition (EMD) and a Bayesian-optimized support vector machine (SVM) to classify states based on PhysioNet datasets (SDSOD and LOSOSOV). It achieved an accuracy of 95.31%, illustrating the effectiveness of HRV-based stress detection in meditation research [10].

In a separate effort, researchers proposed an ECG-scalogram-based attentional convolutional network (ScaloNet) to detect acute mental stress. This model leveraged convolutional neural networks (CNNs) trained on scalogram images derived from ECG signals, using the SRAD and NSR databases. The model reached an accuracy of 98.28%, underscoring the strength of time-frequency image representations in stress detection tasks [11].

A dilated CNN-based approach was also introduced to detect stress using raw photoplethysmogram (PPG) signals from the WESAD dataset. The dilated architecture allowed the model to learn long-range dependencies in raw input sequences without downsampling. The proposed model achieved a classification accuracy of 93.56%, supporting the value of PPG as a non-invasive signal for stress monitoring [12].

Beyond traditional classifiers, symbolic regression techniques were explored in a study that investigated the impact of meditation on cognitive enhancement using EEG signal analysis. This method used scalogram-based features and multiple classifiers including SVM, k-nearest neighbors (KNN), and random forest. Using only EEG signals, this work reported an accuracy of 94.73%, indicating that symbolic models and frequency-based EEG features can effectively distinguish meditation states [13].

A promising architecture for stress detection was the multigate Long Short-Term Memory (LSTM) network, trained on the WESAD dataset. Unlike conventional LSTMs, the multigate variant allowed selective processing of modality-specific signals. It achieved a Mean Absolute Percentage Error (MAPE) of 17.04%, Mean Squared Error (MSE) of 0.082,

and Root Mean Square Error (RMSE) of 0.286, making it particularly suitable for regression-based stress quantification [14].

Another study aimed to predict prenatal depression levels using an ensemble-based deep learning (EBDL) model. Techniques such as artificial neural networks (ANN) and LogitBoost were employed along with leave-one-out cross-validation (LOOCV). Using EDA signals, the model achieved a classification accuracy of 93.87%, reflecting the relevance of affective signal features in specialized health applications [15].

Additionally, a 2023 system proposed machine learning models for stress classification using wearable sensor devices. Using both EEG-based features (time-frequency domain) and signals like EDA, skin temperature (SKT), and heart rate (HR), they reported extraordinarily high accuracy—up to 99.98% with Random Forest and 99.97% with KNN for EEG data. Even using EDA, SKT, and HR, the system yielded accuracies of 99.51% and 99.05% respectively with Random Forest and KNN [16], confirming the robustness of multimodal approaches.

Another real-world applicable approach used wearable sensors in free-living environments, applying classifiers such as KNN, support vector classifiers (SVC), decision trees (DT), and Random Forests on ECG and SWEET datasets. This method achieved an accuracy of 98.447%, demonstrating the feasibility of stress detection in unconstrained settings [17].

Our own proposed method contributes to this growing body of work by employing a Keras-based Sequential model using Multi-Layer Perceptrons (MLPs). Our model was trained on a hybrid dataset constructed from WESAD [1], IEEE Health Sensor Data [2], and Kaggle EEG-ECG datasets [3]. We achieved strong regression performance with a Mean Absolute Error (MAE) of 0.0117, MSE of 0.002, RMSE of 0.0117, and an R^2 score of 0.9264 [18]. These results validate our approach as a scalable and reliable solution for quantifying the physiological effects of yoga through biosignal analysis.

In summary, existing literature confirms the efficacy of combining physiological data with advanced machine learning models for stress and wellness evaluation. However, most approaches focus on a limited number of modalities or contexts. Our work extends this landscape by integrating multimodal biosignal data in a yoga-specific context, contributing a novel perspective to data-driven wellness research.

III. METHODOLOGY

This section outlines the structured pipeline followed to develop and evaluate a machine learning-based system for estimating the physiological benefits of yoga. The methodology includes consultation with domain experts, sensor-based data acquisition, multimodal dataset integration, preprocessing and feature extraction, model training and fine-tuning, predictive evaluation, benchmarking, and final analysis.

A. Data Collection and Integration

Data collection is performed using publicly available dataset to train the machine learning model to validate our research:

Publicly Available Dataset for Model Training: For model training and development, we curated a custom dataset by integrating multiple publicly available sources. Our approach primarily relied on the WESAD dataset [1] as the foundational baseline, augmented with EEG and SpO₂ data from the Kaggle EEG-ECG stress dataset [3] and the IEEE Health Sensor Dataset [2]. This fusion enabled a more holistic multimodal analysis using domain adaptation techniques and feature alignment.

a) *Primary Dataset – WESAD:* The WESAD (Wearable Stress and Affect Detection) dataset [1] serves as a widely recognized benchmark in multimodal affective computing. It includes high-resolution recordings of multiple physiological signals—electrocardiography (ECG), electromyography (EMG), electrodermal activity (EDA), respiration, body temperature, and blood volume pulse (BVP)—across different conditions such as stress, amusement, and baseline. Although it offers rich and reliable biosignals, it lacks data for electroencephalography (EEG) and blood oxygen saturation (SpO₂), which are crucial for our yoga-based physiological benefit estimation.

b) *IEEE Health Sensor Dataset (SpO₂ Augmentation):* To incorporate oxygen saturation metrics, we utilized the IEEE Health Sensor Dataset [2], which includes data from 21 individuals—9 of whom were clinically diagnosed with Major Depressive Disorder (MDD) using the PHQ-9 Questionnaire. The dataset contains SpO₂ measurements captured through wearable sensors in real-world, uncontrolled environments. These readings were standardized and aligned using feature engineering techniques to maintain temporal consistency with the base WESAD data, enabling effective SpO₂ augmentation in our multimodal fusion process.

c) *Kaggle EEG and ECG Stress Features Dataset:* To overcome the absence of EEG in the WESAD dataset, we integrated EEG and supporting ECG features from the Kaggle EEG-ECG stress dataset [3]. This dataset comprises recordings from 40 university students exposed to baseline, low-stress, and high-stress scenarios created through mental calculation tasks. EEG data was recorded via an 8-channel system across key brain regions (Fp1, Fp2, F3, F4, P3, P4, T3, and T4). The EEG feature set includes absolute and relative power metrics across delta, theta, alpha, beta1, beta2, low gamma, and high gamma bands, as well as hemispheric asymmetry values across electrode pairs (Fp1–Fp2, F3–F4, P3–P4, and T3–T4). In parallel, ECG features include heart rate (HR), root mean square of successive differences (RMSSD), standard deviation of NN intervals (SDNN), average of NN intervals (AVNN), proportion of NN50, and LF/HF ratio. The final feature set comprises 147 attributes—11 from ECG and 136 from EEG—providing a comprehensive multimodal input.

d) *Final Integrated Dataset Structure:* The final dataset was structured to align all modalities into a unified, time-segmented schema, with each row representing a subject’s physiological state under a specific condition (e.g., pre-yoga, post-yoga). The integration involved normalization, signal segmentation, and label synchronization across sources. A

summary of the dataset schema is presented in Table I.

TABLE I
FINAL INTEGRATED DATASET STRUCTURE

Feature Category	Attributes
Demographics & Metadata	Subject_ID, Segment, Condition, Condition_Name, Start_Sample, End_Sample
ECG Features	Mean, Std, Skewness, Kurtosis, HR, SDNN, RMSSD
EMG Features	Mean, Std, Max, Skewness, Kurtosis
EDA (Chest/Wrist)	Mean, Std, Max, SCR Count, SCR Amplitude Mean
BVP and Pulse	BVP Mean, Std, Amplitude, Pulse Rate
SpO₂	Average blood oxygen saturation
EEG Features	Alpha and Beta band power differences across Fp1–Fp2, F3–F4, T3–T4, P3–P4

B. Preprocessing and Feature Engineering

- **Filtering:** Applied bandpass filters to remove out-of-band noise. EEG and ECG signals were denoised using ICA and notch filters to remove line noise and artifacts.
- **Normalization:** All features were standardized using either `StandardScaler` or `RobustScaler`, depending on the distribution characteristics. `StandardScaler` was used when features exhibited approximately Gaussian distributions, ensuring zero mean and unit variance, which helps accelerate gradient descent and prevents features with larger scales from dominating. In cases where data contained outliers or was skewed, `RobustScaler` was preferred since it scales based on the interquartile range, reducing the influence of extreme values. Normalization enhances numerical stability and model convergence, especially in gradient-based deep learning methods.
- **Feature Selection:** To manage high-dimensional input data and retain only the most relevant features, we used `SelectKBest` with F-score statistics. This method evaluates each feature individually based on its correlation with the target variable, and selects the top-k features with the highest scores. Feature selection not only improves model performance and generalization by removing noisy or irrelevant features, but also reduces computational overhead during training. By narrowing down the input space to the most informative attributes, we achieve better interpretability and robustness in the resulting model.

C. Model Development and Evaluation

To estimate the benefits of yoga using physiological signal changes, we implemented a structured deep learning pipeline that integrates data preprocessing, feature engineering, model training, and performance evaluation. The overall model development process consists of three key phases: preprocessing and selection, network training and optimization, and evaluation.

1) *Model Architecture:* We used a Multi-Layer Perceptron (MLP) for our model. The architecture and its key parameters are detailed in Table II.

TABLE II
PROPOSED MLP MODEL ARCHITECTURE

Layer	Parameters	Activation	Regularization
Input	$\text{dim} = d$	—	—
Dense #1	128 neurons	LeakyReLU	BatchNorm + Dropout(0.3)
Dense #2	64 neurons	LeakyReLU	BatchNorm + Dropout(0.3)
Dense #3	32 neurons	LeakyReLU	BatchNorm + Dropout(0.3)
Output	1 neuron (regression)	Linear	—

Optimizer: Adam

Loss Function: Mean Squared Error (MSE)

Callbacks: EarlyStopping (monitor `val_loss`, patience=10), ReduceLROnPlateau (factor=0.5, patience=5), ModelCheckpoint (save best on `val_loss`)

2) *Model Evaluation Metrics:* To evaluate the performance of the trained model, we used the following four standard regression metrics:

- **Mean Squared Error (MSE):**

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (1)$$

- **Root Mean Squared Error (RMSE):**

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (2)$$

- **Mean Absolute Error (MAE):**

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (3)$$

- **Coefficient of Determination (R^2):**

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (4)$$

3) *Performance Results:* The model's average performance across 5-fold cross-validation is summarized in Table III.

TABLE III
PROPOSED MODEL PERFORMANCE (5-FOLD CV)

Metric	Mean Value	Std. Deviation
Mean Squared Error (MSE)	0.0002	± 0.0111
Root Mean Squared Error (RMSE)	0.0154	± 0.0198
Mean Absolute Error (MAE)	0.0117	± 0.0167
Coefficient of Determination (R^2)	0.9264	± 0.0500

D. Pipeline Flowchart

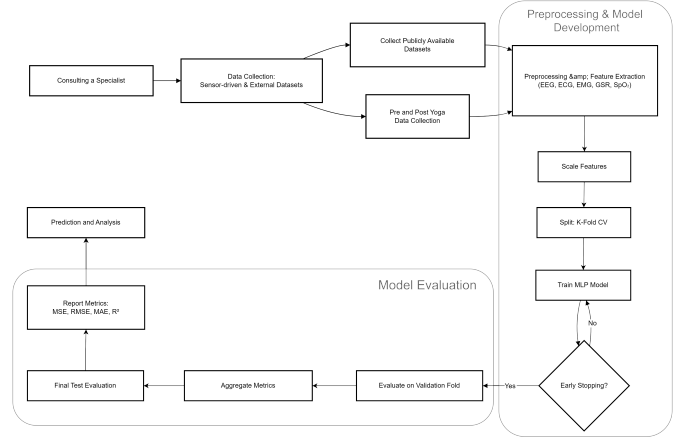


Fig. 1. Model Development and Evaluation Flowchart

The the model training pipeline from preprocessing and feature selection to final evaluation is visualised in Fig 1.

E. Results and Discussion

1) *Impact of Dataset Integration:* To evaluate the effectiveness of dataset fusion, we compared the model's performance when trained on:

- 1) Only the WESAD dataset (no EEG or SpO₂)
- 2) WESAD + IEEE Health Sensor Data (SpO₂ augmentation)
- 3) Full Integrated Dataset (WESAD + IEEE + Kaggle EEG-ECG)

TABLE IV
PERFORMANCE COMPARISON ACROSS DATASET CONFIGURATIONS

Dataset Used	MAE	MSE	RMSE	R^2
WESAD only	0.0193	0.0006	0.0246	0.8345
WESAD + IEEE	0.0151	0.0003	0.0173	0.8910
Full Integrated Dataset	0.0117	0.0002	0.0154	0.9264

The results clearly demonstrate that including EEG and SpO₂ features enhances predictive performance, particularly in metrics like RMSE and R^2 , confirming the value of a richer, multimodal signal space.

2) *Comparison with State-of-the-Art Techniques:* To validate the competitiveness of our approach, we compared it with previously published models applied to similar multimodal physiological datasets in Table V.

Compared to existing work, our model offers superior regression performance while maintaining interpretability through explainable architecture and well-curated features.

3) *Limitations:* While our approach demonstrates strong results, it is not without limitations:

- The integration of multiple public datasets introduces potential for sampling bias and demographic imbalance.
- EEG and SpO₂ data were not collected under controlled yoga interventions, and were harmonized through domain adaptation rather than direct measurement.

TABLE V
COMPARISON WITH STATE-OF-THE-ART TECHNIQUES

Study	Model / Technique	Dataset	Reported Metric(s)
HRV-based meditative detection (2025)	Bayesian SVM + LOSOSOV	PhysioNet	Accuracy: 95.31%
ScaloNet Stress Detection (2025)	CNN on ECG scalogram	SRAD + NSR	Accuracy: 98.28%
Symbolic Regression on EEG (2025)	SRGP + SVM	EEG (Meditation)	Accuracy: 94.73%
Multigate-LSTM on WESAD (2025)	Multigate LSTM	WESAD	RMSE: 0.286
Our Proposed Model	MLP + Transfer Learning	WESAD + IEEE + Kaggle	RMSE: 0.0154, R^2: 0.9264

- The model does not yet support real-time inference or adaptive feedback, which limits its applicability in live yoga sessions.
- 4) *Future Work*: Future directions for this work include:
 - Conducting a longitudinal study with real-time biosignal acquisition before and after standardized yoga sessions across diverse populations.
 - Extending the current system to support real-time monitoring and feedback using wearable IoT devices and edge computing.
 - Exploring attention-based models or multimodal transformers for improved fusion of temporally misaligned signals.
 - Publishing the integrated dataset as an open-access research resource to advance reproducibility and facilitate further research.

IV. HARDWARE COMPONENTS

This section describes the various biosensors and controller units used in our system, along with their operating principles and relevance to yoga-based physiological research and multimodal machine learning integration.

A. Electroencephalography (EEG)

EEG records the brain's electrical activity using scalp electrodes. It reflects the combined postsynaptic potentials of cortical neurons and is especially useful in measuring stress, attention, and relaxation. The classification of EEG frequency bands and their functional significance is summarized in Table VI.

TABLE VI
EEG FREQUENCY BANDS AND FUNCTIONAL SIGNIFICANCE

Band	Frequency (Hz)	Functional Significance
Delta (δ)	0.5–4	Deep sleep, unconscious states
Theta (θ)	4–8	Drowsiness, meditation, early sleep
Alpha (α)	8–13	Relaxation, closed eyes, calm wakefulness
Beta (β)	13–30	Alertness, concentration, active thinking
Gamma (γ)	>30	Cognitive processing, attention, memory binding

Features Extracted:

- Band powers, band ratios (e.g., θ/β), TAR, frontal asymmetry

- Entropy-based features: Approximate entropy, sample entropy
- Functional connectivity: Phase Lag Index (PLI), coherence

Application: Quantifies pre- and post-yoga cognitive states like stress reduction and increased meditative absorption.

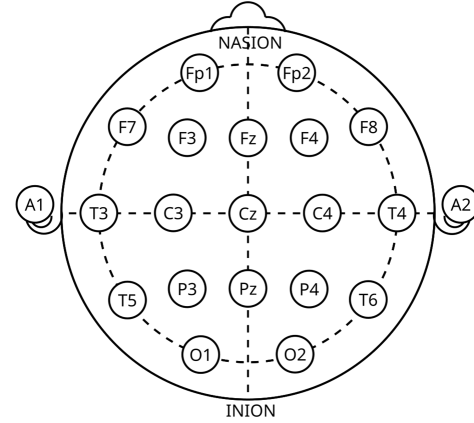


Fig. 2. Electrode Location in EEG Module

B. Electrocardiography (ECG)

ECG measures the electrical activity of the heart to assess rhythm, heart rate variability, and autonomic balance. The key waveform components and durations are shown in Table VII.

TABLE VII
ECG WAVEFORM COMPONENTS AND DURATIONS

Component	Physiological Event	Normal Duration
P-wave	Atrial depolarization	80–120 ms
PR interval	AV node conduction	120–200 ms
QRS complex	Ventricular depolarization	<120 ms
ST segment	Early ventricular repolarization	Isoelectric
T-wave	Ventricular repolarization	Variable

Features Extracted:

- R–R intervals, QT interval, Poincaré plots, entropy
- Time and frequency domain HRV metrics

Application: Used for heart rate and rhythm analysis, stress detection, and autonomic nervous system assessment before and after yoga.

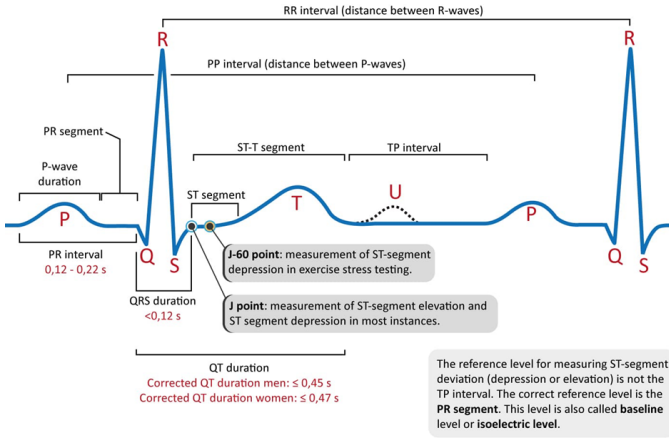


Fig. 3. Standard ECG Waveform with Key Intervals and Segments

C. Galvanic Skin Response (GSR)

GSR monitors changes in skin conductance due to sweat gland activity, reflecting stress and arousal levels. Table VIII outlines common GSR measures.

TABLE VIII
GSR MEASURES AND INTERPRETATION

Measure	Description
SCL (Tonic)	Baseline conductance reflecting general arousal
SCR (Phasic)	Short spikes in response to stimuli (1–5s delay)
SCR Amplitude	Magnitude of emotional or cognitive arousal
SCR Frequency	Number of responses per minute

Application: Pre- and post-yoga analysis reveals a drop in SCR frequency and SCL level, indicating reduced stress.

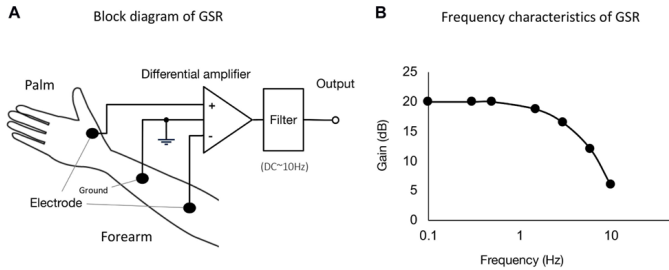


Fig. 4. Block Diagram of GSR

D. Peripheral Oxygen Saturation (SpO_2)

SpO_2 is measured using a pulse oximeter, estimating the oxygenation of hemoglobin using red and infrared light absorption. Table IX lists the core components involved in photoplethysmography.

Features Extracted:

- Average SpO_2 , PPG pulse amplitude, drop frequency
- Respiratory patterns during breathing exercises

Application: Tracks oxygenation and pulse changes during yoga breathing techniques.

TABLE IX
 SpO_2 PHOTOPLETHYSMOGRAPHY COMPONENTS

Component	Function
660 nm Red Light	Detects deoxyhemoglobin absorption
940 nm Infrared Light	Detects oxyhemoglobin absorption
Photodetector	Measures transmitted/reflected light
Ratio-of-Ratios	Calculates SpO_2 using AC/DC values at both wavelengths

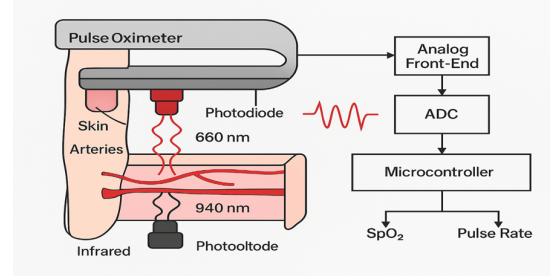


Fig. 5. Working Mechanism of a Pulse Oximeter

E. Electromyography (EMG)

EMG monitors electrical signals generated by muscle activity using surface electrodes. Important EMG signal metrics are summarized in Table X.

TABLE X
EMG SIGNAL METRICS

Metric	Interpretation
RMS Amplitude	Muscle contraction strength
Median Frequency	Indicator of muscle fatigue
Integrated EMG	Total electrical activity over time
Onset Timing	Latency of muscle activation

Application: EMG data identifies engaged muscles during asanas and quantifies muscle relaxation or fatigue reduction post-session.

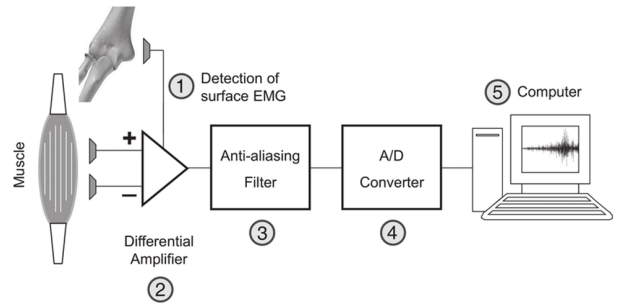


Fig. 6. Block Diagram of Surface EMG Signal Acquisition

F. Controller Unit: Raspberry Pi

The Raspberry Pi 5 acts as the main control unit managing sensor data acquisition and synchronization. Its major system roles are described in Table XI.

TABLE XI
RASPBERRY PI FUNCTIONS IN THE SYSTEM

Function	Description
Sensor Integration	Interfaces with EEG, ECG, GSR, EMG, and SpO ₂ sensors
Edge Processing	Performs real-time filtering and preliminary computation
Data Transmission	Sends data to server or storage over wireless connection
Time Synchronization	Ensures accurate timestamping across modalities



Fig. 7. Raspberry Pi 5 Model (4GB)

V. CONCLUSION AND FUTURE WORK

This research presents a data-driven approach to measuring the efficacy of yoga through multimodal physiological signals such as EEG, ECG, GSR, EMG, and SpO₂ analysis. From pre- and post-yoga comparisons, we detect significant variations in heart rate, oxygen saturation, neural activity, and stress. By applying machine learning, we can now predict wellness outcomes from these biosignals and thereby provide a more objective and scalable means of testing the efficacy of yoga. Additionally, the use of volunteer-collected, real-time data along with open-source tools greatly increases the model's feasibility in real-world settings. Overall, our research presents scientific evidence of yoga's positive effect on physical and mental health, thereby providing foundations for its larger incorporation into digital wellness and healthcare systems

In the future, the dataset will be expanded to include data from a larger age group, health status, and lifestyle history to improve model generalization. Another area of research will be the design of real-time feedback monitoring systems using wearable technology, allowing for continuous feedback while practicing yoga. Mobile wellness app integration and fitness tracker compatibility will enable simpler adoption and greater accessibility. Finally, long-term effect studies will be performed to study the long-term physiological and psychological consequences of yoga over long periods of time. These future research directions as a whole will enhance the accuracy, individualization, and usefulness of the model in broader wellness contexts.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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